

QUANTUM-INSPIRED APPROACH TO DETECT SQL- INJECTION ATTACK

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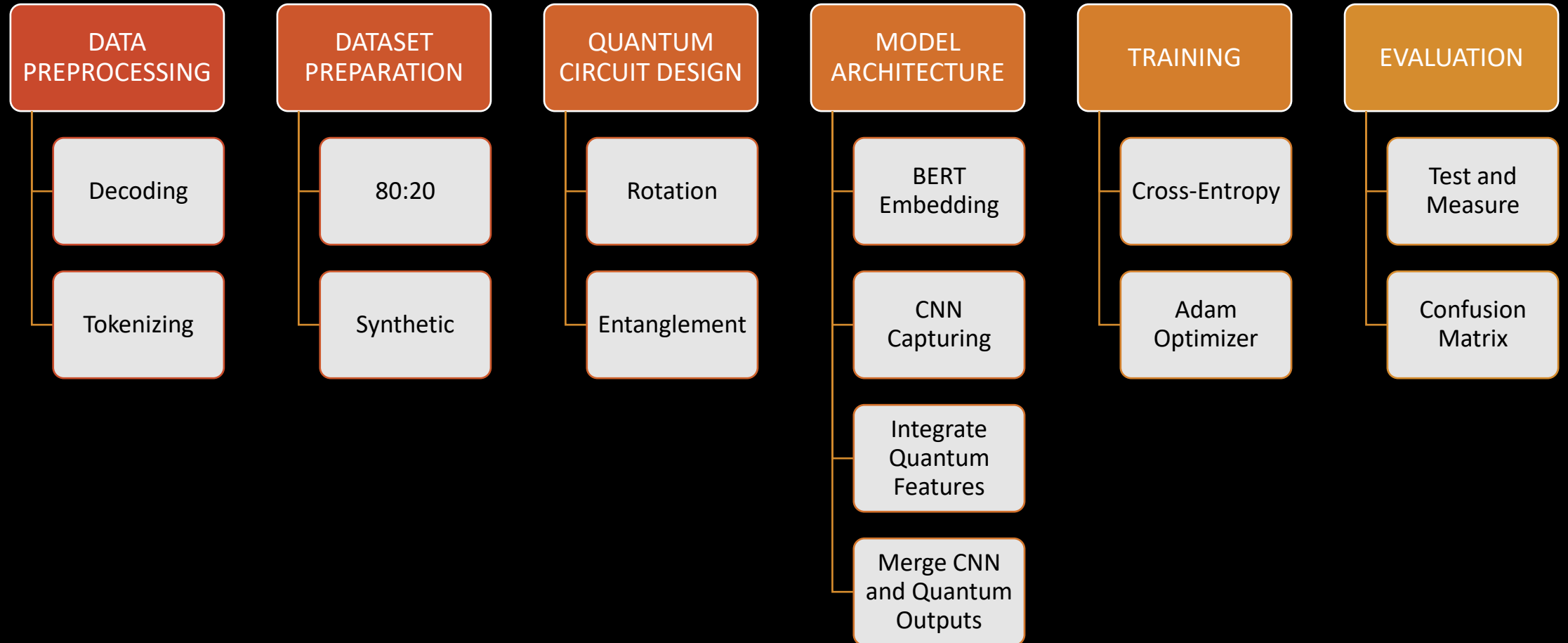
WHY QUANTUM APPROACH?

- Quantum Encoding
- Non-linear patterns
- Bias – Variance Tradeoff

PROBLEM STATEMENT & PROPOSED SOLUTION

- SQL injection (SQLi) is a serious web security flaw.
- Traditional methods fail to detect new and complex SQLi attacks.
- Challenges: poor scalability, high computation, and limited adaptability.
- ***Quantum-Inspired Machine Learning (QiML) offers better scalability, efficiency, and detection of advanced SQLi patterns.***

PROPOSED ARCHITECTURE



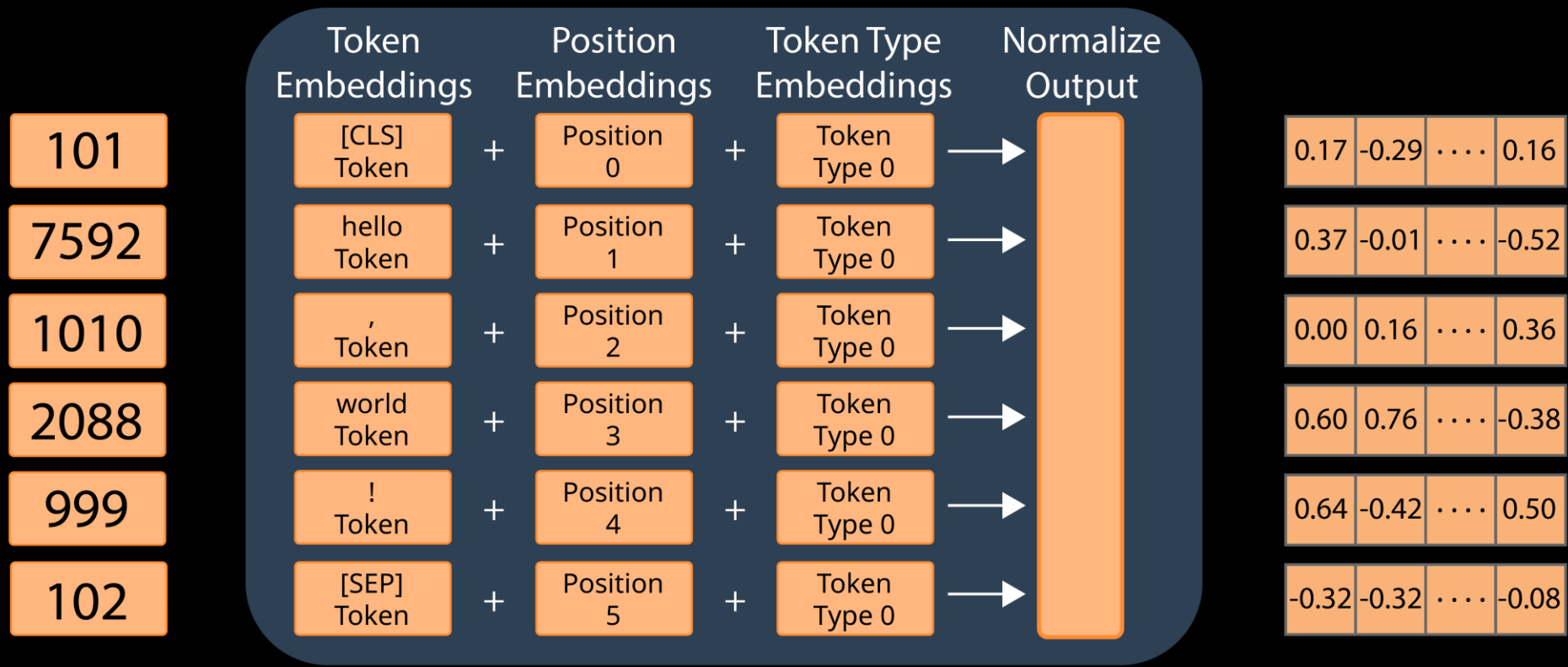
QUANTUM CIRCUIT DESIGN

- **Rx Rotation:** The Rx gate rotates a qubit around the X-axis of the Bloch sphere by an angle θ .
- **Ry Rotation:** The Ry gate rotates a qubit around the Y-axis of the Bloch sphere by an angle θ .
- **Entanglement** is a quantum phenomenon where the states of multiple qubits become interdependent such that the state of one qubit cannot be described independently of the others.

BERT EMBEDDING

- BERT embeddings play a critical role in SQL injection detection by transforming input text (e.g., SQL queries) into high-dimensional semantic representations. These embeddings capture contextual nuances, enabling machine learning models to:
 - Understand Syntax and Semantics
 - Detecting Anomalies
 - Enhancing Accuracy

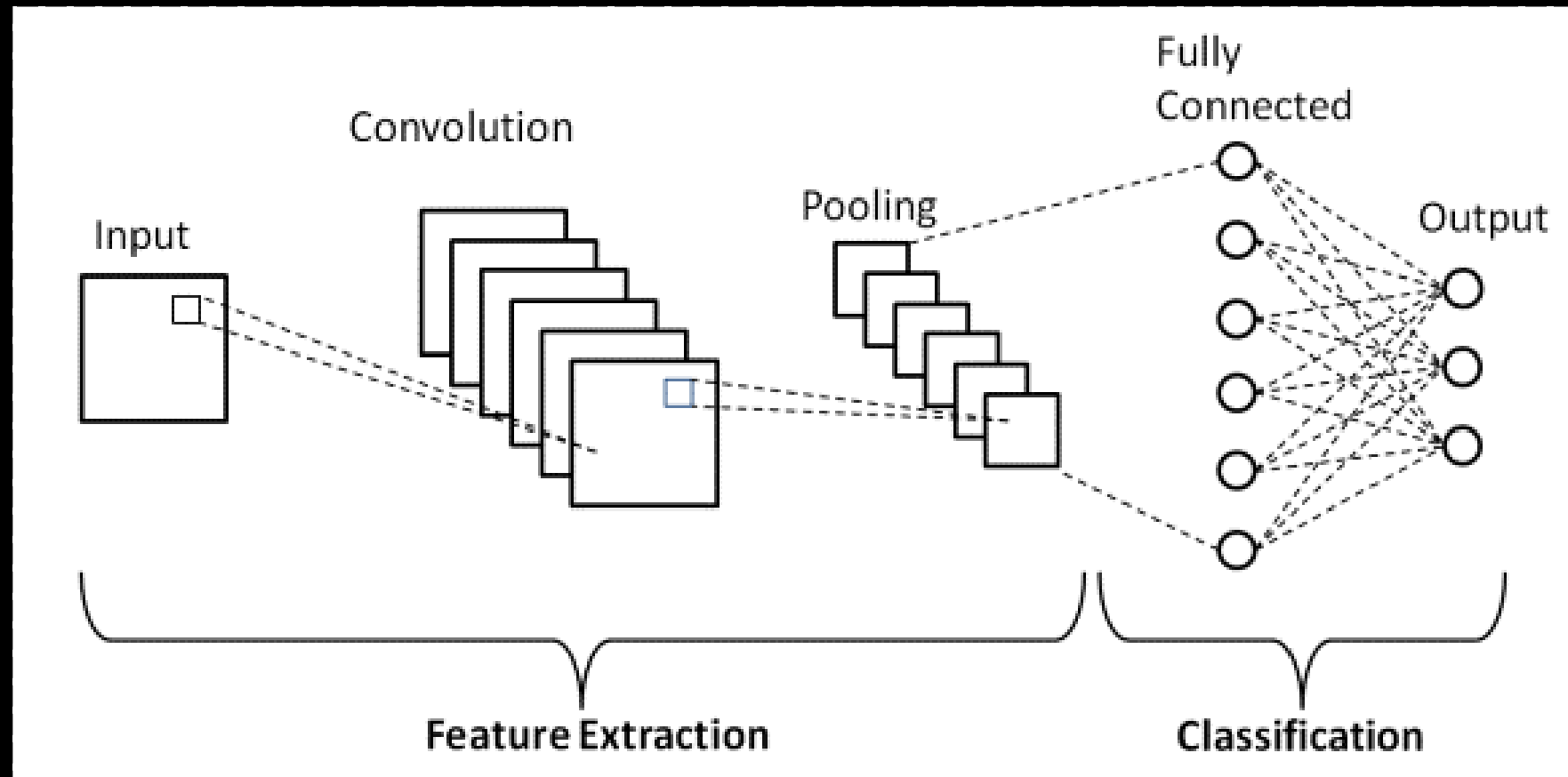
ARCHITECTURE OF BERT



CNN CAPTURING

- Convolutional Neural Networks (CNNs) play a vital role in SQL injection detection by capturing spatial and sequential patterns in input queries.
 - Pattern Recognition
 - Feature Extraction
 - Scalability

ARCHITECTURE OF CNN



SUMMARY

- Model 1: Detecting SQL Injection Attacks using Classical Machine Learning Techniques (CML)
- Model 2: Detection SQL Injection Attacks using Quantum-Inspired Machine Learning Techniques (QiML)
- Dataset:
 - Modified_SQL_Dataset.csv
 - Shape: (30920, 2)
 - Features: Query, Label

RESULTS

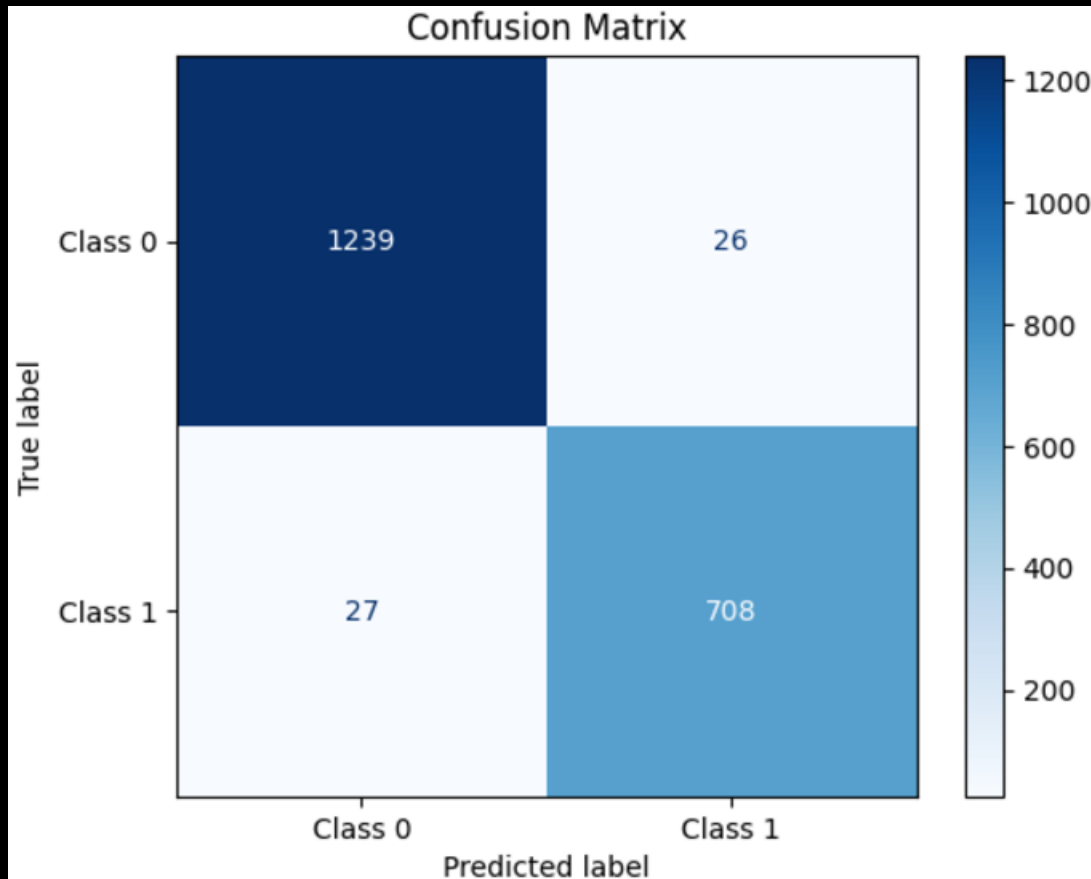
- Metrics: Accuracy, Recall, Precision, F1-score

Parameters	Model 1 (CML)	Model 2 (QiML)
Batch Accuracy*	91.8%	87.5%
Test Accuracy	97.35%	97.50%
Precision	96.46%	98.77%
Recall	96.33%	94.65%
F1-Score	96.39%	96.67%

* Indicates the accuracy for training these models over a single epoch

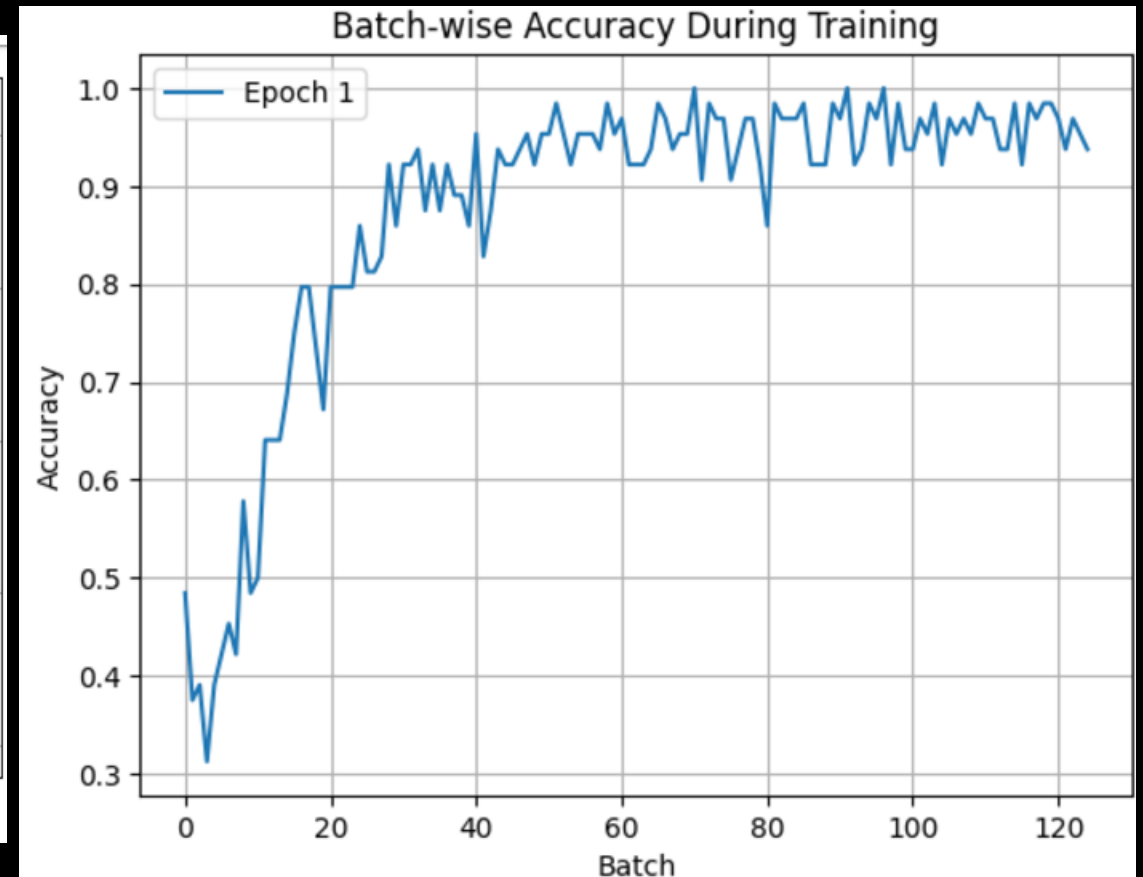
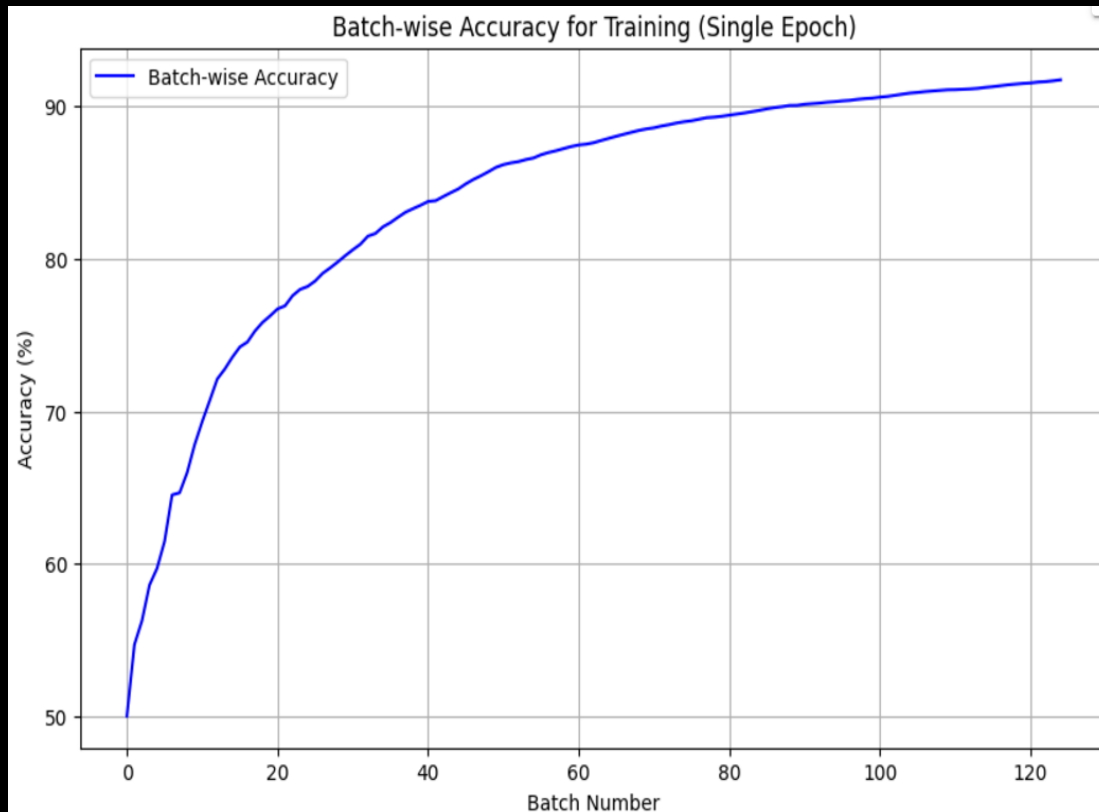
RESULTS

- Confusion Matrices for CML (left) and QiML (right):



RESULTS

- Performance Graphs for CML (left) and QiML (right):



CONCLUSION

- Model 2 (QiML) performs slightly better overall due to higher test accuracy, precision, and F1-score, thus making it more effective in minimizing false positives.
- However, Model 1 (CML) excels in recall and batch accuracy, making it better at detecting more SQL injection attacks during training.
- The choice depends on whether precision or recall is prioritized.